



Quantum Oncology: The Applications of Quantum Computing in Cancer Research

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Abstract A global technological race is underway to develop increasingly powerful and precise quantum computers. As a transformative computing paradigm, quantum computing offers the potential for exponentially accelerating specific algorithms, thereby providing the necessary computational power to process vast amounts of data. In light of the challenges classical computing faces with the complexity and volume of oncology data, we introduce the concept of “quantum oncology” and explore its potential applications throughout the cancer care continuum. Additionally, we address several challenges and potential solutions for integrating quantum computing into oncology research. By illuminating these issues, we aim to deepen the understanding of quantum computing’s potential in oncology and advocate for multidisciplinary collaboration to propel the advancement of precision oncology.

Clinical Trial Number Not applicable.

Keywords Quantum computing · Oncology · Quantum machine learning · Cancer detection · Cancer diagnosis · Quantum radiotherapy · Anticancer drug development

Abbreviations

EMR	Electronic medical record
AI	Artificial intelligence
ML	Machine learning
QML	Quantum machine learning
Q-Oncology	Quantum oncology
GQC	Gate-based quantum computers
QA	Quantum annealing
CT	Computed tomography
ART	Adaptive radiotherapy
IMRT	Intensity-modulated radiotherapy
qDRL	Quantum deep reinforcement learning
CADD	Computer-aided drug design
KRAS	Kirsten rats arcomaviral oncogene homolog

NISQ

Noisy intermediate-scale quantum

Background

Cancer continues to pose major social, public health, and economic challenges globally. Recent statistics indicate nearly 20 million new cancer cases in 2022, with 9.7 million cancer-related deaths, and these numbers are expected to increase in the coming decades [1]. Despite considerable progress in diagnosing and treating cancer over the past decade, tumor heterogeneity leads to significant inter-individual variations in treatment response and clinical outcomes. To enhance the understanding of inter- and intra-tumour heterogeneity, recent significant advancements in high-throughput diverse omics technologies have propelled precision oncology into an era of “big data” [2]. The digital transformation of clinical oncology, including the enhancement of electronic medical record (EMR) systems and the digitization of radiology and pathology imaging, has also increased the volume and complexity of patient data [3]. The exponential growth in oncology data presents formidable challenges for computational power and storage capacity. Traditional computers struggle to process this data swiftly and effectively or to develop personalized treatment

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strategies promptly. This raises a crucial question: What strategies can be implemented to surmount these limitations and enable efficient data analysis? Quantum computing, with its emerging parallel processing and simulation capabilities, promises to be a future solution to the computational challenges of oncology.

Quantum computing is a novel computing paradigm that leverages quantum mechanical laws and properties to manipulate quantum information units for computation [4]. Since its initial proposal in the 1980s, the field of quantum computing has experienced significant fluctuations [5]. It has advanced remarkably, transitioning from academic exploration to practical applications over the past four decades (Fig. 1). Shor's factorization algorithm leverages the Fourier transform to efficiently identify periodicities, thereby reducing the problem of integer factorization to one solvable in polynomial time. Its discovery catalyzed early efforts to construct quantum computers [6], sparking extensive research into quantum algorithms. More recently, in 2019, researchers conducted the first successful quantum supremacy experiment on a 53 qubit superconducting machine, which completed a task in 200 s that would take traditional computers 10,000 years [7]. Owing to its quantum advantage, quantum computing has attracted significant interest as a potential next-generation technology, with the 2020s heralded as a critical period for quantum technology development [8]. Recent developments have enabled the exploration of potential quantum advantages in healthcare and life sciences [9, 10]. Notably, in 2023, IBM's Quantum System One, the world's first quantum computer dedicated

to healthcare research, was unveiled, marking a new dawn for healthcare and public health [11].

Precision medicine fundamentally involves analyzing multiscale and multimodal data from a broad spectrum of patient-contributed sources to customize personalized treatments for individuals and enhance patient outcomes [12]. Artificial intelligence (AI), especially machine learning (ML), is designed to extract valuable insights from datasets and develop predictive models for decision-making [13]. From this perspective, AI proves to be a potent tool for achieving precision medicine [14, 15]. In oncology research, AI is increasingly integrated into various clinical "touchpoints" along the cancer care pathway, including risk prediction, screening, diagnosis, grading, staging, prognosis, treatment response assessment, and follow up [15, 16]. In an era of growing convergence between big data and AI, the perspective of incorporating mathematical optimization problem into cancer care pathways will further elucidate the promise of AI in clinical oncology applications [16]. However, classical ML algorithms face inherent limitations in processing large datasets or solving combinatorial optimization problems. Quantum AI, or quantum machine learning (QML), represents a big advancement in data analytics, merging quantum computing principles with ML techniques to tackle complex data analysis challenges [17]. The synergy of quantum computing, AI, and big data holds immense potential to accelerate the evolution of precision oncology [2, 18].

Given the escalating challenges of scale and complexity that classical computing faces in addressing the sheer

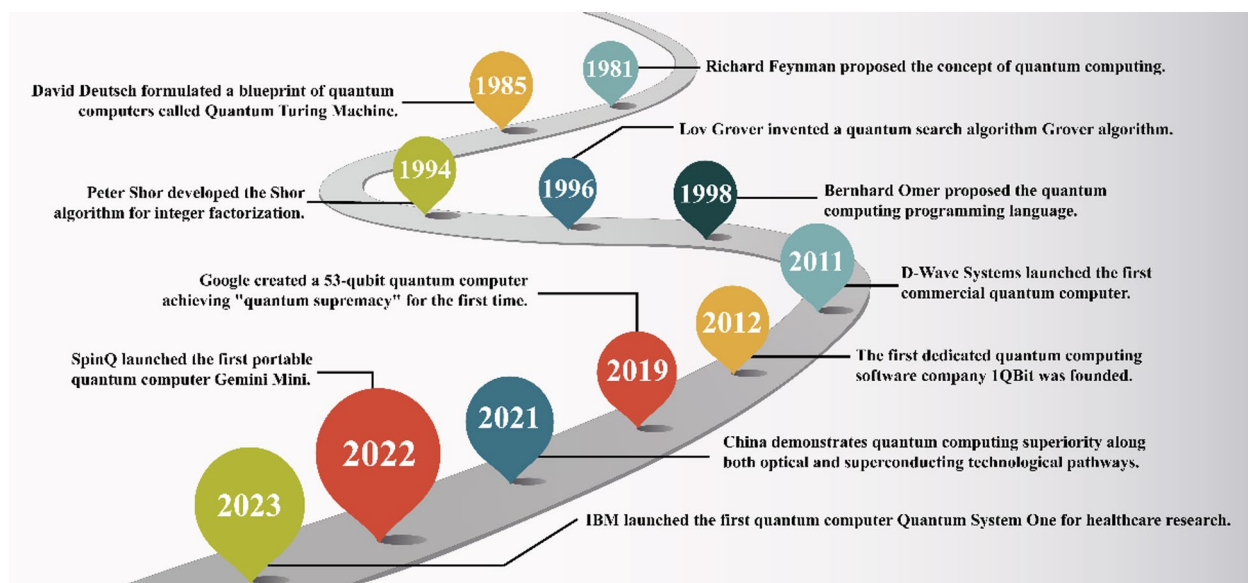


Fig. 1 A timeline depicting the evolutionary path of quantum computing over the past four decades. This spans from the initial proposal of the quantum computing concept to the introduction of the first commercial quantum computer and its early applications in healthcare,

marking quantum computing's steady transition from theoretical to practical applications. In oncology, quantum computing is not only viewed as having substantial potential but is also considered a likely direction for future developments

volume and variety of big data in oncology, we introduce the concept of Quantum Oncology (Q-Oncology). This approach integrates quantum technologies, cancer biology, clinical oncology, and computational sciences, with a specific focus on leveraging quantum computing for large-scale data analysis, discovery of therapeutic targets, drug development, and optimization of cancer treatment pathways. It aims to surpass the bottlenecks in data processing and analysis encountered in traditional oncology, enhancing the precision, efficiency, and personalization of cancer research and treatment. In this review, we first provide a brief overview of the principles of quantum computing, then detail the application of quantum computing in oncology across various touchpoints along the cancer care pathway, and finally, we describe the challenges inherent in the practical application of quantum computing. By exploring these issues in the context of rapid advances in quantum computing, we aim to foster concepts that will drive meaningful research and ultimately translate into real-world clinical applications.

Quantum Computing Fundamentals

The primary motivation for introducing quantum computing is to address the performance bottleneck of classical computers. Instead of conventional binary bits, quantum computing leverages qubits, which can simultaneously represent multiple states. This property enables the parallel processing of large-scale data in real time [19]. A qubit, a two-dimensional quantum system, exists in a superposition of the $|0\rangle$ and $|1\rangle$ logic states, equivalent to the binary values of 0 and 1. The superposition state of a qubit can be described as the following information-theoretic representation: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where α and β are complex numbers that satisfy the condition $|\alpha|^2 + |\beta|^2 = 1$. The Bloch sphere, an essential visualization tool, maps all possible quantum states onto a unit sphere's surface [20] (Fig. 2a). Measurements play a critical role in quantum systems, maintaining the state's integrity. On measurement, a qubit's superposition collapses to either 0 with a probability of $|\alpha|^2$ or 1 with a probability of $|\beta|^2$ [21].

Additionally, quantum entanglement and quantum interference are pivotal for quantum computing. Quantum entanglement ensures that the states of two or more qubits remain interconnected, even over large distances. Altering one qubit instantaneously modifies the state of its entangled counterpart, facilitating effective information transmission between them [22]. Quantum interference, stemming from the wave-particle duality of quantum particles, occurs when different quantum states overlap (Fig. 2b). Constructive interference enhances the wave function when phases align, whereas destructive interference cancels it out when they diverge significantly [23]. This phenomenon is essential

for executing many quantum algorithms, as constructive interference increases the likelihood of the correct solution, while destructive interference minimizes errors, directing the system towards the optimal outcome.

Quantum computing adheres to the basic computational stages of input, processing, and output (Fig. 2c). Loading data is a critical initial step in the quantum computing process. When tackling real-world oncology problems, it's crucial to first encode classical information such as genomic, clinical, imaging, and pathological data into qubit states through state preparation. This facilitates further manipulation and processing on a quantum computer [24]. Gate-based quantum computers (GQC) then manipulate these states in a quantum register using a series of unitary transformations based on quantum gates. These transformations, central to quantum computation, lead to an output that, after a separate measurement step, reveals the computation results [25]. Given their inherent probabilistic nature, quantum algorithms must be run multiple times to produce a results distribution, from which the most frequent outcome is selected.

While GQC offers a universal framework for solving diverse problems, quantum annealing (QA) provides a specialized alternative for optimization issues [26]. QA uses quantum tunneling to escape local minima and identify global optima in complex energy landscapes [27]. Unlike GQC, QA continuously evolves the quantum system to efficiently search for solutions in specific optimization problems, particularly useful in combinatorial optimization [28], making it a potentially superior choice for complex challenges in oncology.

Quantum Computing Applications and Touchpoints along Cancer Treatment Pathways

Advancements in oncology research have moved from low to high-dimensional data analyses, driven by more complex data generation and computational technology improvements. This shift opens new avenues for understanding cancer heterogeneity. Quantum computing, surpassing conventional computing limits, shows considerable potential to address critical cancer care challenges through its parallel processing and precise simulation capabilities [29]. It's crucial to approach quantum computing development in oncology from both patient and clinician perspectives, focusing on essential touchpoints such as tumor screening, early detection, diagnosis, classification, staging, prognosis prediction, treatment response, radiotherapy optimization, and omics analyses (Fig. 3). Integrating quantum computing into these areas can deepen the analysis of complex patient data, uncover potential biomarkers, and create innovative,

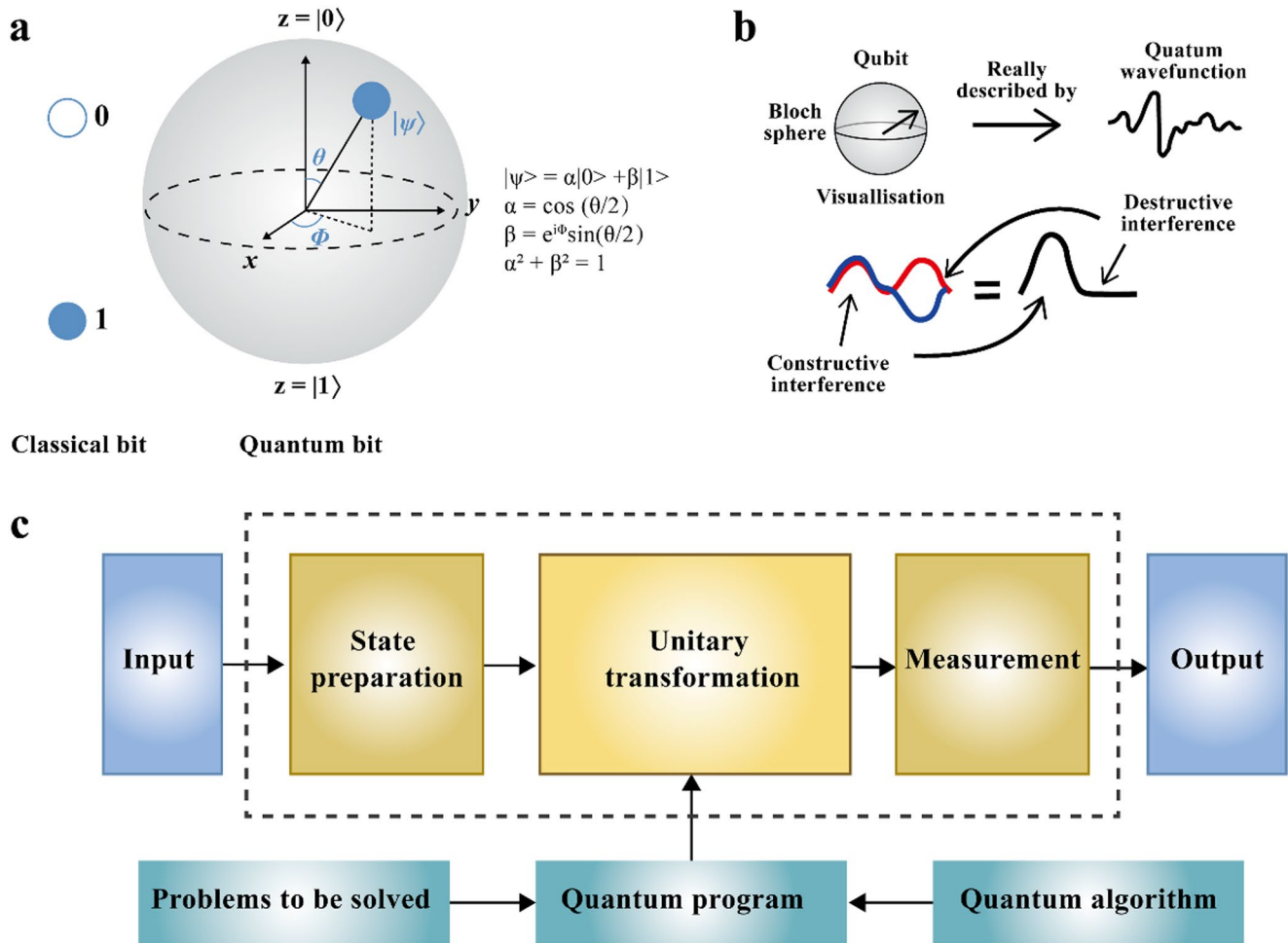


Fig. 2 Fundamental concepts and structures in quantum computing. **(a)** A qubit can exist in a superposition of 0 and 1 states, unlike classical bits. The Bloch sphere offers a visualization of a qubit's continuous states. **(b)** Quantum interference, which utilizes the wave-particle duality of particles, enables constructive interference to amplify

waves, while destructive interference nullifies them. **(c)** Classical data is initially encoded into quantum states via a state preparation process. An appropriate quantum algorithm is then applied for unitary transformation, tailored to the problem at hand, culminating in a measurement to derive the computational results

personalized treatment strategies, enhancing the precision and effectiveness of cancer therapies.

Tumour Screening and Early Detection

Tumour screening and early detection in healthy populations are vital for reducing cancer morbidity and mortality by identifying asymptomatic cases and intervening at pre-cancerous stages. Tumour screening data, derived from serum markers, imaging, and endoscopy, benefit from quantum computing's ability to rapidly process diverse data sources, potentially improving early detection efficiency and accuracy. The earliest research combining quantum computing and cancer detection technologies dates back a decade, with improvements like the Shor algorithm enhancing cancer detection over traditional methods [30]. With the advancement of quantum computing technology, increasingly

sophisticated QML techniques, including kernel-based and quantum neural network-based approaches, are being applied to cancer detection. Notable examples are the integration of quantum transfer learning with tumour markers such as CA-125, HE4, and ROMA, which has demonstrated promising results in assessing ovarian cancer risk [31], and the deployment of quantum convolutional neural network models for endoscopic image analysis aimed at early esophageal cancer detection [32].

Conventional imaging methods such as X-ray and low-dose computed tomography (CT) have been enhanced by quantum computing, which not only improves the accuracy of detecting cancer in these images [33] but also refines tumor boundaries and intensifies tumor features through advanced edge detection [34–36] and effective image noise reduction [37], thereby aiding clinicians in precise diagnosis. Given the current heavy workload in cancer screening,

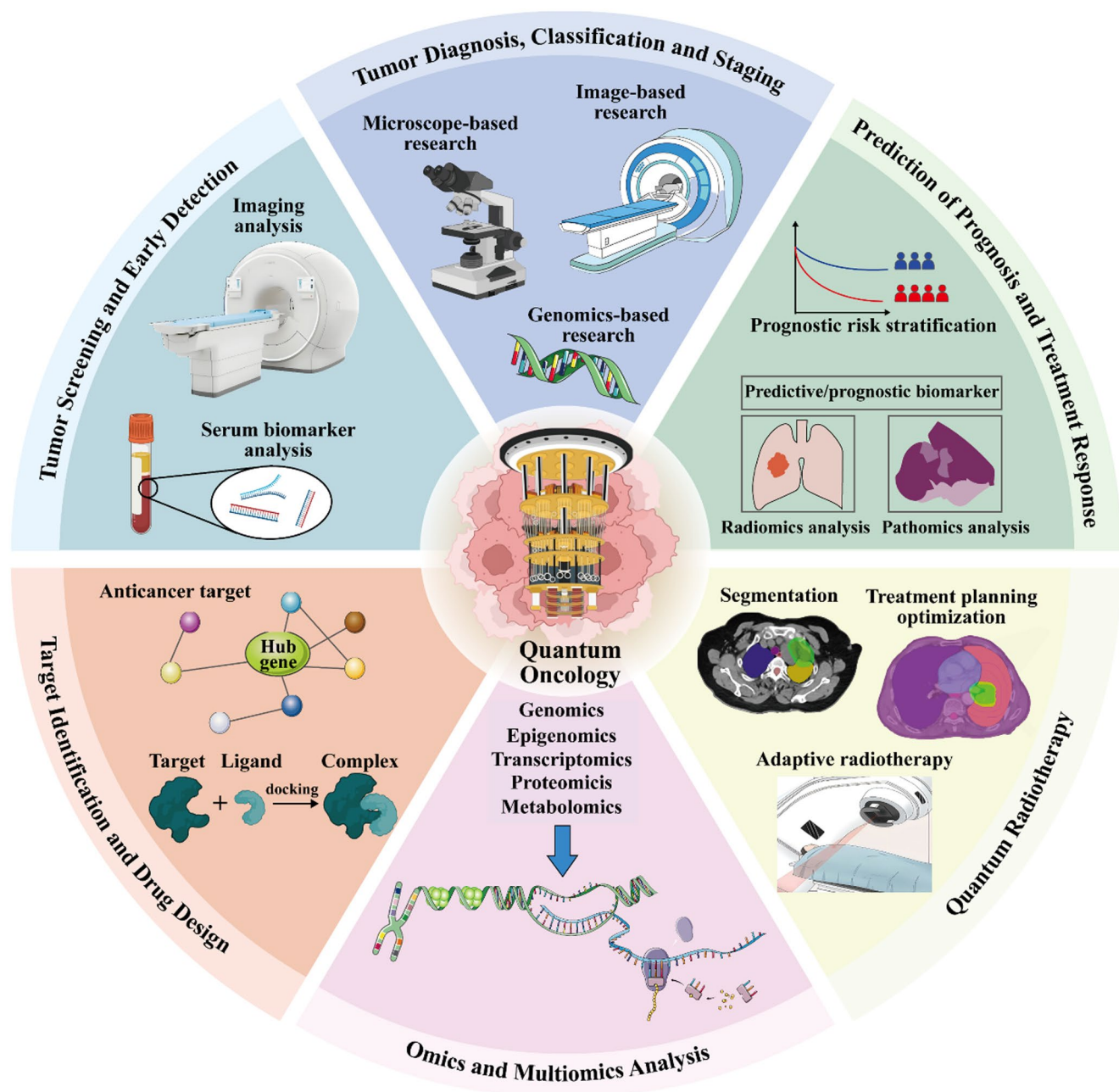


Fig. 3 Principal applications of quantum computing in oncology. Capitalizing on its capabilities for handling large-scale data and optimizing complex systems, quantum computing facilitates new avenues for

early detection, accurate diagnosis, personalized treatments, and the creation of innovative anticancer drugs, thus propelling the progress of precision oncology

quantum computing is poised to enhance the efficiency and reduce the burden on radiologists. In addition, quantum sensors, due to their high sensitivity, play a crucial role in early cancer detection. For example, Saren et al. [38] have shown that quantum dot-based biosensors can detect tumour-specific proteins associated with gastrointestinal tumours, such as CEA, CA125, CA19-9, and AFP, at very low levels in the blood. However, the extensive and diverse multivariate data generated by quantum sensors poses analytical challenges.

Quantum computing, by processing this data in real time, offers the potential for early cancer detection and ongoing personalised treatment guidance [39].

Tumour Diagnosis, Classification, and Staging

Accurate diagnosis, classification, and staging are key to determining tumor aggressiveness and significantly impact clinical decisions and treatment outcomes. Cancer diagnosis

integrates information from histopathology, imaging, and increasingly, genomic data. Based on the current test data, quantum computing has been shown to diagnose cancer, classify cancer subtypes, and determine cancer stages more accurately than traditional computing methods by analyzing data from histopathology, radiology (e.g., CT and magnetic resonance imaging), and genomics [40–46]. In the realm of cancer diagnosis and classification, researchers begin by compiling extensive sample data with binary or multiclass labels (e.g., malignant or benign), labeled as “basic facts”, and assess quantum computing’s performance across various metrics including accuracy, specificity, sensitivity, and the area under the receiver operating characteristic curve. Comparisons of quantum computing’s performance with that of classical computing and across different quantum settings (e.g., quantum simulators versus quantum computers) are integral to these studies [47]. However, some studies have not observed quantum computing’s superiority due to current limitations in the number of quantum bits and quantum hardware [48]. To overcome these challenges, researchers have employed various optimization methods, such as nature-inspired algorithms [49] and adaptive machine learning [50], to boost the efficiency and performance of quantum circuits. These methods increased processing speeds, reduced model complexity, and improved accuracy and noise resistance, thus enhancing precision in complex data analysis. For cancer staging, quantum algorithms not only offer higher efficiency in data analysis but also provide information on lymph node status and metastasis compared to traditional methods [51]. Ultimately, quantum computing enables more accurate diagnosis and staging, facilitating a comprehensive characterization of a patient’s disease state and providing critical guidance for subsequent clinical decisions.

Prediction of Prognosis and Treatment Response

Prognosis and treatment response are critical indicators for assessing the outcomes of cancer patients, reflecting disease progression and treatment efficacy. Therefore, accurately predicting these outcomes is vital for medical management. In this research area, developing reliable predictive models and effective predictive markers are key to achieving accurate predictions. A robust model not only enhances the accuracy and reliability of predictions but also expands their clinical applicability. Yu et al. [52] demonstrated the potential of quantum computing to improve predictive prognostic modeling, which has shown to enhance performance over traditional methods. Importantly, this study utilized a real-world, small-sample dataset, highlighting the potential of quantum algorithms to expand the analytical space in cases with limited data, such as rare diseases. In addition,

preliminary studies suggest that quantum computing could improve drug response prediction accuracy compared to classical algorithms in specific scenarios, but further validation is required [53]. Quantum computing can also model the complex dynamics of tumor burden, considering the natural growth rate of the tumor and the drug-induced contraction rate during treatment [54]. As we enter the era of computer vision, computers are increasingly capable of extracting high-throughput information about tumour heterogeneity from radiographic images and histopathologic sections, using techniques such as radiomics and pathomics. By integrating these techniques, quantum computing can more rapidly and accurately identify potential predictive and prognostic markers from vast amounts of radiographic and histopathologic data. For example, the integration of quantum computing with radiomics has enabled the prediction of 3-year and 2-year survival rates for lung cancer and glioma patients, respectively, with error mitigation techniques further enhancing the robustness of these quantum predictive models [55].

Quantum Radiotherapy

Radiotherapy, a critical component of cancer treatment, involves approximately 50–70% of patients receiving it as part of their regimen [56]. This complex, multi-step process, from pre-radiation treatment decisions to post-radiation follow-up, requires collaboration among a multidisciplinary team including radiation oncologists, medical physicists, dosimetrists, and radiation therapists. As radiotherapy technology rapidly advances, the number and complexity of variables and parameters involved in the treatment process have significantly increased, necessitating support from human-machine interaction modes for decision-making. However, classical computers often lack the computational efficiency and accuracy required for managing the vast variables and parameters, complicating the demand for real-time adjustment and multidimensional optimization. Quantum computing, with its developing parallel processing capabilities, may offer advantages in addressing certain challenges in radiotherapy optimization [57]. Several application scenarios of quantum computing in the radiotherapy process have been documented in the literature, including treatment planning optimization, adaptive radiotherapy (ART), and image segmentation [58, 60, 63].

The first known application of quantum computing in radiotherapy emerged in 2015, exploring the use of QA for optimizing intensity-modulated radiotherapy (IMRT) treatment plans [58]. This initial study focused on two prostate cancer cases, comparing QA’s effectiveness against two conventional optimization methods: simulated annealing and Tabu search. Although QA’s performance matched that

of the Tabu search, it did not reach the efficacy of simulated annealing. However, QA was three to four times faster than both methods. This finding suggests that QA-inspired methods could surpass traditional clinical optimization techniques in speed as QA hardware evolves. Subsequent research applied quantum tunneling annealing, which leverages the quantum tunneling effect, to enhance IMRT optimization for two liver cancer cases. This study confirmed improvements in convergence speed and robustness in treatment planning quality [59]. Variations in patients' genetic and physiological characteristics can influence their response to radiotherapy, necessitating personalized treatment plans tailored to individual circumstances. Niraula et al. [60] developed a novel quantum deep reinforcement learning (qDRL) framework by integrating reinforcement learning algorithms with quantum computing and deep learning to optimize decision-making in ART for lung cancer. By simulating the treatment process and incorporating quantum state decision-making mechanisms, the framework progressively refined the treatment plan within an artificial radiotherapy environment influenced by biological, physical, genetic, clinical, and dosimetric factors, aiming to enhance treatment outcomes and minimize side effects. Analyses indicated that the qDRL-based framework could potentially increase radiotherapy efficacy by at least 10%. Tumor shrinkage or daily anatomical changes during radiotherapy can lead to geometric shifts in patient anatomy, potentially rendering initial treatment plans suboptimal or obsolete. Therefore, developing real-time, efficient predictive models to anticipate patient anatomical changes is crucial for supporting ART decision-making. Pakela et al. [61] proposed a novel method to predict patient changes during treatment by modeling their anatomical shifts as a stable quantum system. Although based on an approach analogous to quantum physics, this represents a promising direction for applying quantum computing in ART. As an area closely related to medical imaging, the segmentation of tumors and organs at risk is essential in radiotherapy. Previously noted benefits of quantum computing, such as enhanced edge detection and image noise reduction [34–37], suggest that segmentation models incorporating quantum computing can more effectively manage complex and heterogeneous medical images [62]. Quantum walking exploits quantum superposition to explore all possible paths in parallel, and then filters out the effective paths by quantum interference. Its application in image segmentation reduces the workload of manual annotation, enhancing segmentation efficiency and accuracy, and decrease the dimensionality required for quantum computation [63]. In summary, quantum computing offers new computational paradigms to explore directions for radiation therapy optimisation, though clinical

translation and measurable impact on treatment outcomes remain to be demonstrated.

Omics and Multiomics Analysis

The advent of various omics technologies, such as genomics, epigenomics, transcriptomics, proteomics, and metabolomics, has provided deeper insights into the molecular behavior of tumors at multiple levels. Integrated multi-omics data analysis can create a comprehensive cancer atlas, elucidate complex interaction networks, and reveal tumour plasticity and heterogeneity. Quantum computing offers substantial advantages in processing the vast volume and multidimensional information of multiomics data. Reports in the literature have documented quantum acceleration in single-omics analysis, such as DNA sequence alignment via quantum gate algorithms [64] and single-cell RNA sequencing utilizing QA [65]. Additionally, quantum computing's application in multi-omics research is showing promising potential. Li et al. [66] demonstrated the use of a QA-based Ising model for the first time in genome-wide multi-omics human cancer data analysis. This study categorized various cancer types, such as breast, colorectal, and lung cancer, along with their molecular subtypes, using data from The Cancer Genome Atlas. The results indicated that the Ising model performed comparably to traditional algorithms yet excelled on smaller data training sets, corroborating findings from Yu et al. [52]. The implementation of quantum computing in multi-omics analysis is anticipated to refine the accuracy of tumor type and molecular subtype classification [66], optimize treatment options [67], and advance personalized treatment development.

Target Identification and Drug Design

Drug discovery is a time-consuming and costly process, spanning target identification and validation to final clinical application. While computer-aided drug design (CADD) has somewhat accelerated this process, it remains limited by computing power and resource constraints. Quantum computing shows promise in enhancing drug discovery and development by potentially accelerating data processing and molecular simulations.

At the outset of new drug development, target identification is critical. Leveraging its quantum advantage, quantum computing efficiently parses multilevel omics data, reconstructs tissue-specific networks, and elucidates interactions across molecular levels. It provides a crucial foundation for identifying novel anti-cancer targets within extensive search spaces [68]. Quantum computing addresses two significant hurdles: computing reaction barriers and simulating molecular dynamics in drug design. Lau et al. [69] have combined

quantum with classical computing to create a hybrid workflow, HypaCADD, managing computationally demanding tasks such as molecular docking, binding affinity prediction, and molecular dynamics simulations. This underscores the vital role of quantum computing in enhancing the CADD process. Utilizing quantum mechanics/molecular mechanics simulations further deepens our understanding of drug-target interactions, with applications in researching covalent inhibitors targeting kirsten rats arcomaviral oncogene homolog (KRAS), a pivotal target in cancer therapy [70]. Additionally, quantum generative models, integrated with quantum computing, show promise in synthesizing new KRAS inhibitors, offering advantages over traditional models [71]. The use of quantum chemical calculations in developing anti-cancer drugs also marks a research direction for applying quantum computing in drug development [72, 73]. The prospects and challenges of enhancing quantum chemical calculations with quantum computing have been explored in a recent review [74]. In conclusion, quantum computing shows wide-ranging application prospects throughout the stages of the drug design process, from target identification and ligand synthesis to complex formation.

Challenges and Future Perspectives

Although quantum computing presents immense potential in oncology, it remains nascent compared to established classical computing. The fragility of quantum hardware, software development limitations, and computational resource scarcity significantly restrict its application in this field. Future research must address these challenges to fully exploit quantum computing in cancer detection, diagnosis, treatment, and fundamental research, facilitating advancements in personalized medicine and precision oncology.

Decoherence and Quantum Error Correction Techniques

Quantum bits are notoriously fragile, susceptible to external noise and environmental changes, which erode quantum state coherence. This decoherence induces computation errors, thus impairing the precision of simulations and calculations, and poses a significant challenge in the current noisy intermediate-scale quantum (NISQ) system. Quantum error correction techniques are considered essential for mitigating decoherence and are a primary focus of current research. Advances in these techniques are progressing us towards a resilient, large-scale quantum computer.

Data Security and Anti-Quantum Computing Cryptography Systems

Data security is a critical concern in protecting the privacy and addressing ethical issues associated with tumor patient data. The evolution of quantum computing threatens existing network and data security systems; for example, the Shor algorithm can decipher RSA encryption in polynomial time [75]. To counteract quantum computing's impact on cryptographic systems, research is focused on developing post-quantum cryptography [76] and quantum secure communication, utilizing quantum key distribution [77]. Implementing anti-quantum computing cryptographic systems is anticipated to safeguard oncology data in the quantum era.

Multidisciplinary Collaboration and Quantum Training

The development of quantum computers and their application in oncology necessitate close collaboration and deep integration across multiple disciplines, including physics, materials science, computer science, clinical oncology, bioinformatics, and cancer biology. Although quantum computing has demonstrate initial success in certain areas, its principles and applications remain largely unfamiliar to most oncology researchers. Consequently, specialized training in quantum computing is crucial; it not only enables oncologists to comprehend the application of quantum algorithms in cancer detection, diagnosis, and treatment optimization, but also motivates them to design quantum algorithms that are more aligned with oncology needs, thus facilitating their direct applications in clinical practice.

Conclusion

In conclusion, quantum computing, as an emerging technology, offers promising prospects for oncology research and may contribute to advancements in precision oncology. This article has reviewed and highlighted several promising applications of quantum computing in oncology research, including cancer detection, diagnosis, treatment optimization, and anti-cancer drug development. However, the application of quantum computing in oncology is still hindered by challenges such as decoherence, scalability, and insufficient interdisciplinary collaboration. Future research should aim to overcome these technical and practical barriers to maximize the use of quantum computing in translating theoretical research into practical oncology applications.

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Author Contributions SHY designed the study, edited and approved

the final manuscript. LL and NL collected the materials. TH and FHX analyzed the materials and graphed the figures. JRL analyzed the materials and drafted the manuscript. All authors read and approved the final manuscript.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Ethics Approval and Consent to Participate Not applicable.

Consent for Publication Not applicable.

Competing Interests The authors declare no competing interests.

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